Inquisitive semantics beyond questions

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Inquisitive semantics: Introduction

- Inquisitive semantics is a semantic framework for the analysis of linguistic information exchange.
- Information exchange is seen as a process of raising and resolving issues.
- Inquisitive semantics provides a new formal notion of issues, which makes it possible to model the semantic content of both declarative and interrogative sentences in an integrated way, capturing not only the information that such sentences convey but also the issues that they raise, which makes it possible to model various concepts that are crucial for the analysis of linguistic information exchange in a more refined and more principled way than has been possible in previous frameworks.

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Inquisitive semantics: Motivation

Why do we need a formal notion of issues?

Reason 1: To represent the content of interrogative sentences

- The semantic content of a declarative sentence is standardly construed as a set of possible worlds, those worlds that are compatible with the information that the sentence conveys.
- This set of worlds is referred to as the proposition that the sentence expresses. This notion of semantic content works well for declarative sentences, whose main conversational role is indeed to provide information.
- Can the semantic content of an interrogative sentence be construed as a set of possible worlds as well?

(1) Ivan is calling.

(2) Is Ivan calling?

- "An interrogative sentence and an indicative one contain the same thought; but the indicative contains something else as well, namely, the assertion. The interrogative sentence contains something more too, namely a request. Therefore two things must be distinguished in an indicative sentence: the content, which it has in common with the corresponding sentence-question, and the assertion." (Frege, 1918)
- So the idea is that declaratives and interrogatives have the same semantic content—a proposition—but come with a different force—either assertion or request.

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Inquisitive semantics: Motivation

Why do we need a formal notion of issues?

Reason 1: To represent the content of interrogative sentences

(1) Is Ivan calling, or Penny?
(2) Who is calling?
(3) John knows that Ivan is calling.
(4) John knows whether Ivan is calling.

- If the embedded clauses had the same content, then by the principle of compositionality the two sentences as a whole should also have the same content. But this is clearly not the case. So the embedded clauses must differ in content.
- Thus, the standard notion of semantic content does not seem applicable to interrogative sentences. Rather, what we need for interrogatives is a notion of content that directly captures the issues that they raise.

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Inquisitive semantics: Motivation

Why do we need a formal notion of issues?

Reason 2: To model conversational contexts

Conversational contexts are modeled in a way that does not only take account of the information that has been established in the conversation so far, but also of the issues that have been brought up, often referred to as the questions under discussion.

(1) A: What did you do this morning?

B: I read the newspaper. Implicature: B did not do the laundry

(2) A: What did you read this morning?

B: I read the newspaper. Implicature: B did not read anything besides the newspaper

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Inquisitive semantics: Motivation

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(1) A: Who did Alf rescue?

B: Alf rescued ZOE [FOC]. / #ALF [FOC] rescued Zoe.

(2) A: Who rescued Zoe?

B: ALF [FOC] rescued Zoe. / #Alf rescued ZOE [FOC].

Focus marking, just like pragmatic reasoning, is sensitive to the issue under discussion.

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Inquisitive semantics: Motivation

Why do we need a formal notion of issues?

Reason 3: To model issue-directed propositional attitudes and capture the meaning of verbs that report such attitudes

Just like there are verbs like *know* and *believe* that describe the information state of an agent, as in (1) below, there are also verbs like *wonder* and *be curious* that describe the inquisitive state of an agent, as in (2).

(1) John knows that Bill is coming.

(2) John wonders who is coming.

Knowledge and belief are information-directed propositional attitudes; wondering is an issue-directed propositional attitude. The standard way to model the knowledge and beliefs of an agent is as a set of possible worlds, namely those worlds that are compatible with what the agent knows or believes. Such a set of worlds is thought of as representing the agent's information state. Similarly, in order to capture what an agent wonders about, we need a representation of her inquisitive state. For such a representation, we again need a formal notion of issues.

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Inquisitive semantics: Motivation

Declaratives and interrogatives can’t be treated separately

Reason 1: Mutual embedding

Declarative and interrogative sentences can be embedded into one another:

(1) Bill asked me who won.	embedded interrogative
(2) Who told you that Jane won?	embedded declarative
(3) Bill asked me who told you that Jane won.	two-level embedding

So the meaning of a declarative sentence is sometimes partly determined by the meaning of an embedded interrogative sentence, and vice versa.

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Inquisitive semantics: Motivation

Declaratives and interrogatives can’t be treated separately

Reason 2: Interpretational dependencies

The interpretation of a declarative sentence depends on the issue raised by a preceding interrogative:

(1) A: What did you do this morning?

B: I only read the newspaper. Implicature: B did not do the laundry

(2) A: What did you read this morning?

B: I only read the newspaper. Impl.: B did not read anything besides the newspaper

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Inquisitive semantics: Motivation

Why do we need an integrated notion of semantic content?

Reason 1: Common building blocks

Declaratives and interrogatives are to a large extent built up from the same lexical, morphological, and intonational elements.

(1) Luca is from Italy↑ or from Spain↓.

(2) Is Luca from Italy↑ or from Spain↓?

In declaratives, *or* is normally taken to yield the union of the semantic values of the disjuncts. In (1), each disjunct expresses a proposition, standardly represented as a set of possible worlds: the semantic value of the first disjunct is the set of worlds where Luca is from Italy, and the semantic value of the second disjunct is the set of worlds where Luca is from Spain. The proposition expressed by (1) is the union of these two sets. This seems a reasonable account of or in declaratives. But what is the role of or in interrogatives?

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Inquisitive semantics: Motivation

Why do we need an integrated notion of semantic content?

Reason 2: Entailment

We need a notion of entailment which is sensitive to both informative and inquisitive strength

(1) a. The number of planets is 8. Entails:

b. The number of planets is even.

(2) a. What is the number of planets? Entails:

b. Is the number of planets even?

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Inquisitive semantics: Motivation

Why do we need an integrated notion of semantic content?

Reason 3: Logical operations

(1) Peter rented a car and Mary booked a hotel.

(2) Peter rented a car or he borrowed one.

(3) Where can we rent a car, and which hotel should we take?

(4) Where can we rent a car, or who might have one that we could borrow?

(5) If Bill asks Mary out, she will accept.

(6) If Bill asks Mary out, will she accept?

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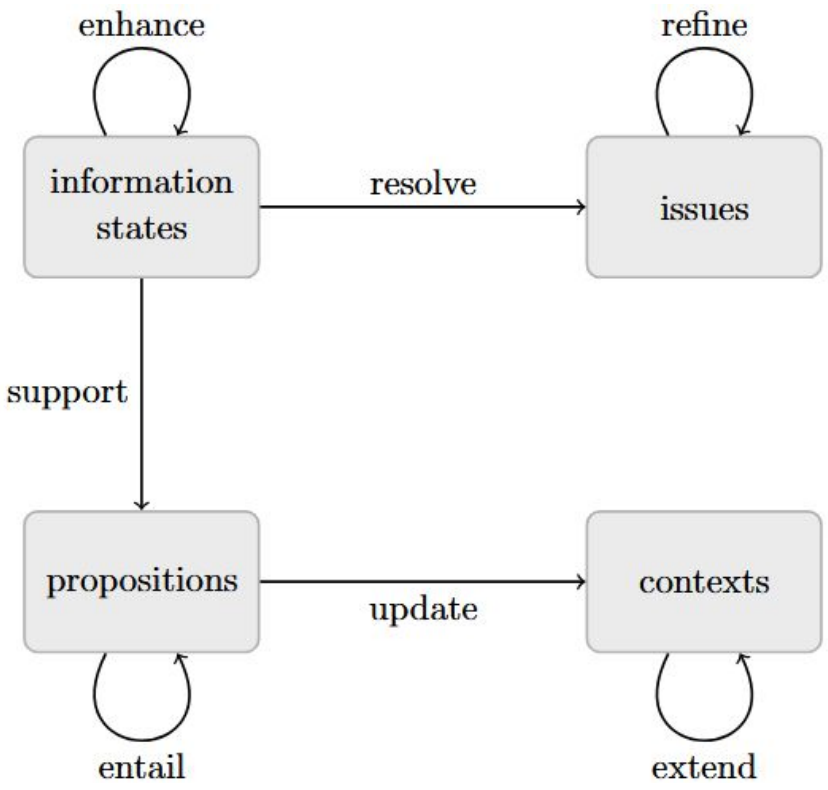
Inquisitive semantics: Basic notions



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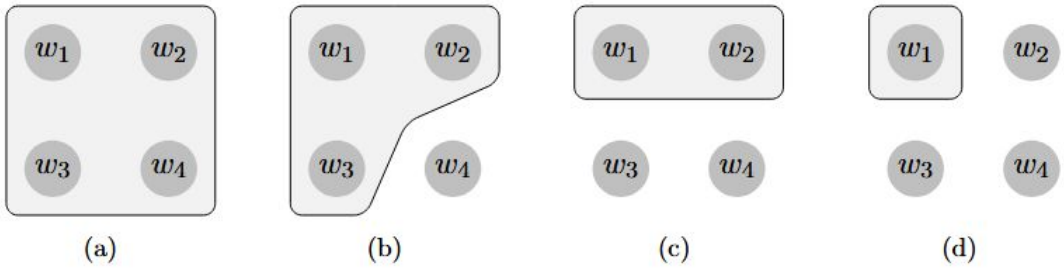
Inquisitive semantics: Information States

- Information states are modeled as sets of possible worlds—those worlds that are compatible with the information available in the state.
- An information state s is a set of possible worlds, s ⊆ W, where W denotes the set of all possible worlds.
- An information state can be thought of as locating the actual world within a certain region of the logical space. The state in (d) contains the information that the actual world is located in the upper left corner of the logical space.



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Inquisitive semantics: Information States

- An information state *s* is a set of possible worlds, s ⊆ W, where W denotes the set of all possible worlds.
- A state t is called an enhancement of s just in case t ⊆ s.
- Is the state (b) is an enhancement of the state in (a)?

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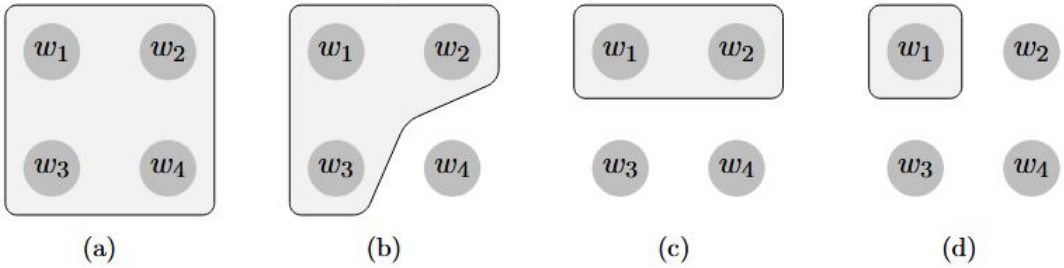

Inquisitive semantics: Information States

- The state (a) is referred to as the ignorant state. Every other state is an enhancement of it.
- At the other far end of the enhancement order is the empty state, ∅. This is an enhancement of any other state. It is a state in which all possible worlds have been discarded as candidates for the actual world, i.e., the available information has become inconsistent. It is therefore referred to as the inconsistent state.

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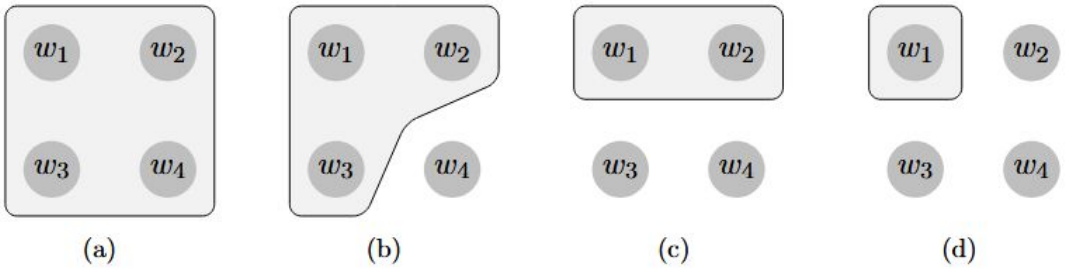
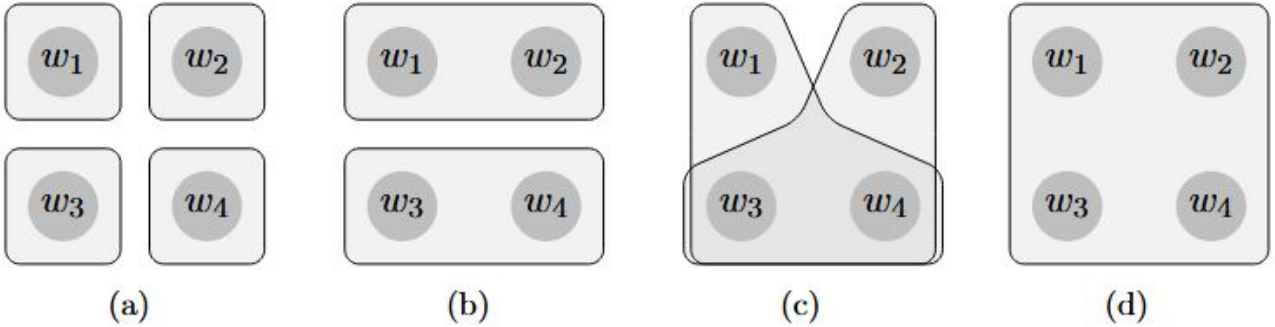


Inquisitive semantics: Issues

- An issue is a non-empty, downward closed set of information states.
- We say that an information state s resolves an issue I just in case s $\in$ I. If s resolves I, we will sometimes also say that I is settled in s.
- The issue depicted in subfigure (a) can only be settled consistently by specifying precisely which world is the actual one.



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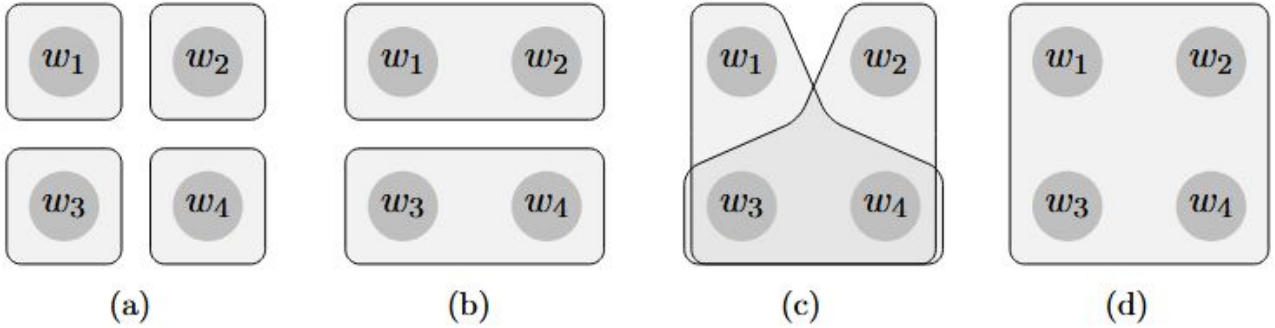
Inquisitive semantics: Issues

- Let I be an issue and s an information state. Then we say that I is an issue over s if and only if ∪I = s.
- Let I, J be two issues over a state s. Then I is at least as inquisitive as J if and only if I ⊆ J. In this case, we say that I is a refinement of J.
- Which issue below is the most inquisitive issue over W, and thus a refinement of all other issues? Which issue below is the least inquisitive issue over W, and all other issues are refinements of it?



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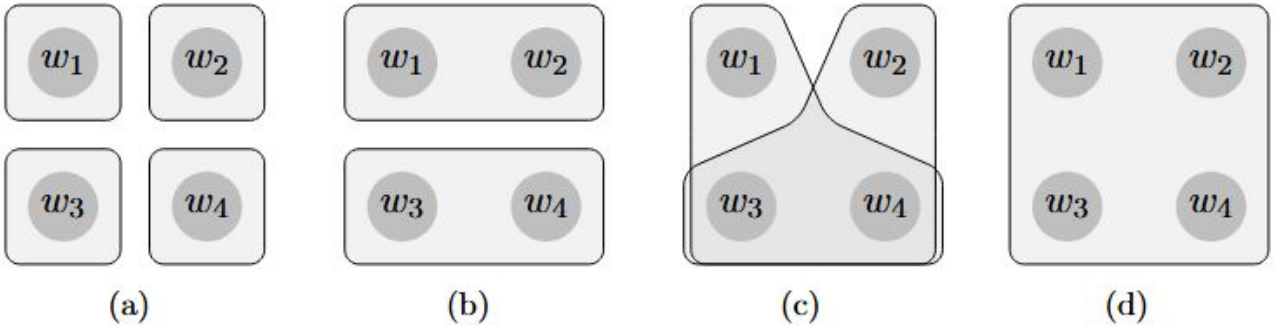
Inquisitive semantics: Issues

- The maximal elements of an issue I are called the alternatives in I.
- Which of the issues below are not inquisitive (contain a single alternative)?
- Which of the issues are inquisitive (contain multiple alternatives)?



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Inquisitive semantics: Propositions

- A proposition P is a non-empty, downward closed set of information states.
- For any proposition P: informative content info(P) := ∪P
- The issue embodied by a proposition P is the one that is resolved in a state s just in case s ∈ P .
- A proposition P is true in a world w just in case w ∈ info(P).
- An information state s supports a proposition P if and only if s ∈ P .
- A proposition P is true in a world w if and only if P is supported by {w}.
- The informative content of a proposition is the set of all worlds in which the proposition is true.

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Inquisitive semantics: Propositions

- We will say that a proposition P is informative just in case its informative content is non-trivial, i.e., $\text{info}(P) \neq W$.
- We will say that P is inquisitive just in case establishing its informative content is not sufficient to settle the issue that it raises, i.e., $\text{info}(P) \notin P$.
- The maximal elements of a proposition P are called the alternatives in P.
- The set of alternatives in P is denoted as alt(P).
- A proposition containing finitely many elements is inquisitive if and only if it contains multiple alternatives.

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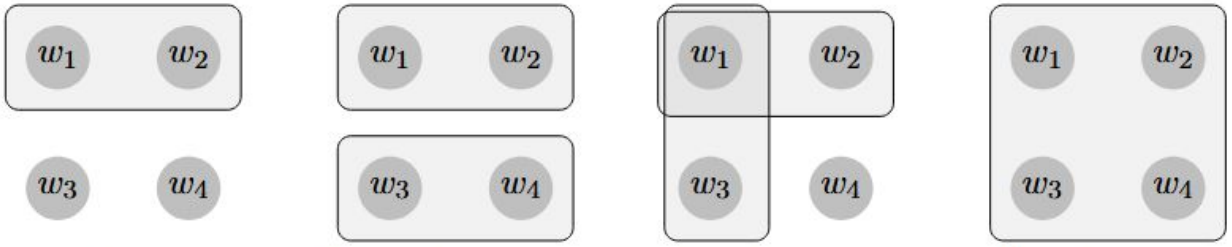
Inquisitive semantics: Propositions

- P is non-inquisitive iff info(P) $\in$ P .
- P is non-informative iff info(P) = W .
- P is a tautology iff W $\in$ P.
- Which one of the propositions below is a tautology?



(a) +informative −inquisitive (b) −informative +inquisitive (c) +informative +inquisitive (d) −informative −inquisitive

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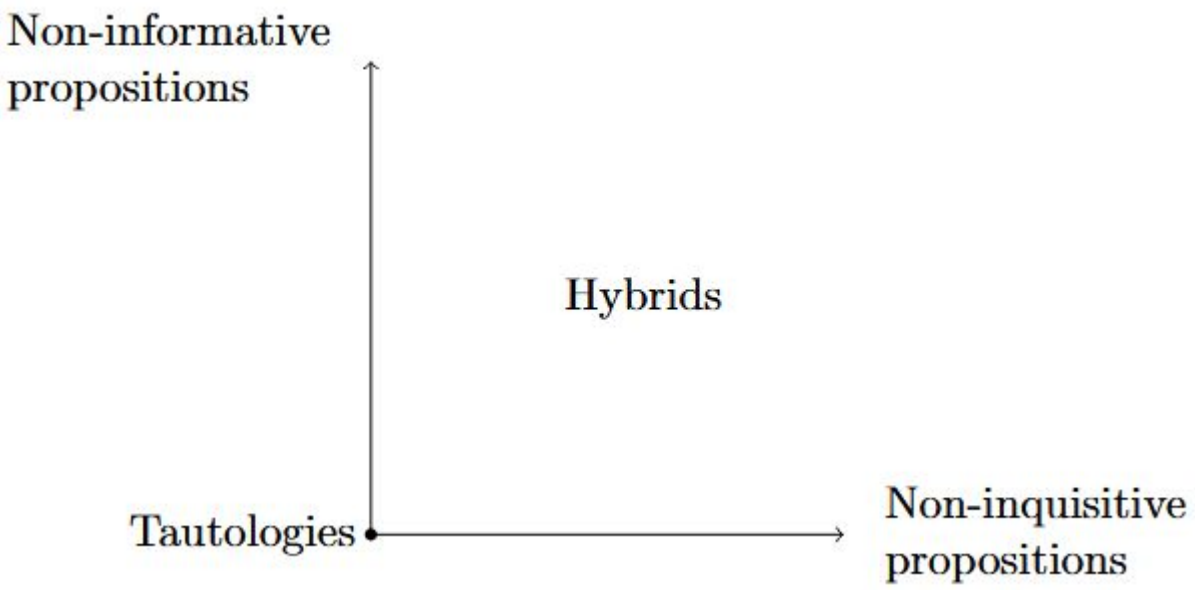
Inquisitive semantics: Propositions



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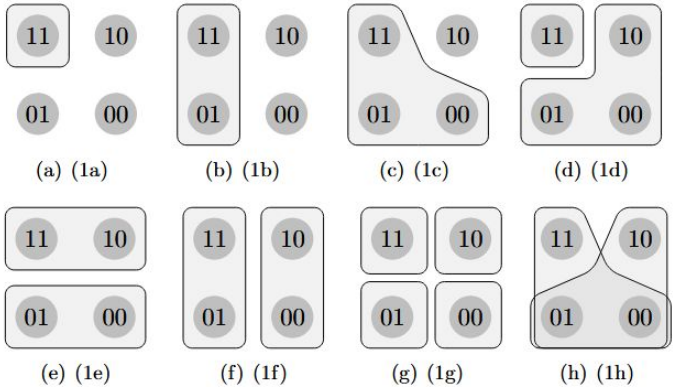
Inquisitive semantics: Propositions

- P entails Q just in case P is both at least as informative and at least as inquisitive as Q.

(1) a. The code is 11.
b. The second digit of the code is 1.
c. If the first digit is 1, the second digit is also 1.
d. Is the code 11?
e. What is the first digit?
f. What is the second digit?
g. What is the code?
h. If the first digit is 1, what is the second?



- While the semantic contents of (1a)-(1c) could have been captured just as well by means of the standard notion of a proposition as a set of possible worlds, the enriched notion of propositions allows us to analyze also the interrogatives in (1d)-(1h). In each case, the relevant proposition is inquisitive, since it contains multiple alternatives, and it is not informative, since these alternatives jointly cover the entire logical space. This captures the fact that, in uttering one of (1d)-(1h), a speaker raises an issue and does not provide any information.

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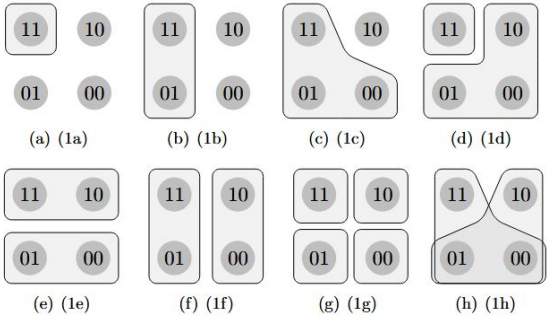
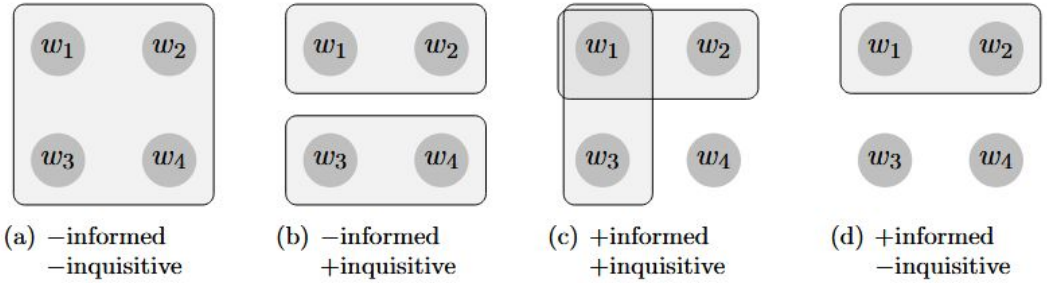


- (1a) entails (1b), which in turn entails (1c).
- For questions in (1d)-(1h) informative content is trivial, and entailment will compare inquisitive strength; an entailment between a pair of questions will hold if the issue expressed by the first is at least as demanding as the issue expressed by the second.
- Does (1g) entail any of the other questions?
- Does (1f) entail (1h)?
- Questions in (1d) and (1e) are incomparable in terms of inquisitive strength: on the one hand, the information that the code is not 11 resolves (1d) but not (1e); on the other hand, the information that the first digit is 1 resolves (1e) but not (1d); thus, neither of these questions entails the other.
- A declarative entails an interrogative just in case the information provided by the former suffices to resolve the issue raised by the latter.
- (1a) entails all of the above questions; (1b) entails only (1f) and (1h); and (1c) entails only (1h).

Inquisitive semantics: Contexts

- A context C is a non-empty, downward closed set of information states.
- For any context C: info(C) := ∪C
- A context C is informed iff info(C) ⊋ W.
- A context C is uninformed iff info(C) = W.
- A context C is inquisitive iff info(C) ∉ C.
- A context C is indifferent iff info(C) ∈ C.



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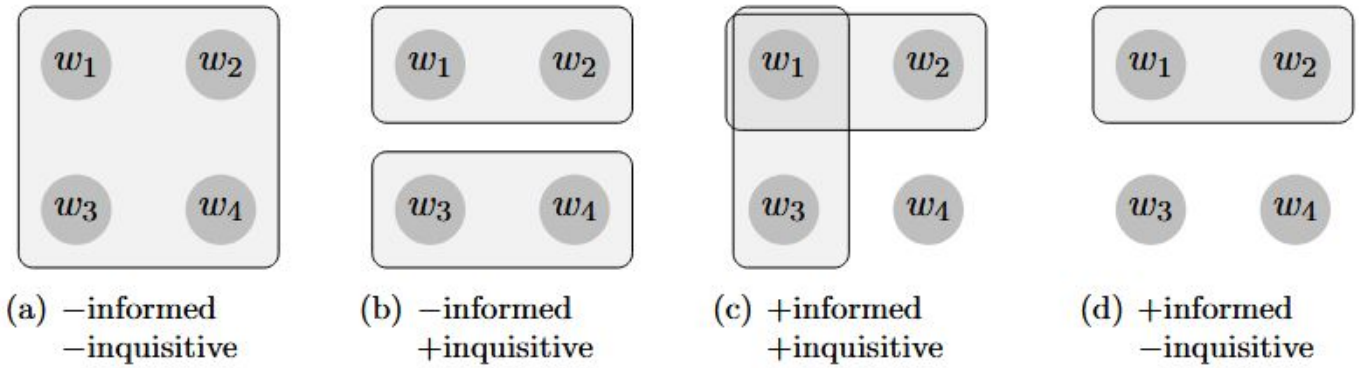
Inquisitive semantics: Contexts

- C′ is an extension of C, C′ ≥ C, iff C′ ⊆ C.
- The contexts in (b), (c), and (d) are all extensions of the trivial context in (a).
- (d) is also an extension of (b) and (c), but neither (b) nor (c) is an extension of the other.



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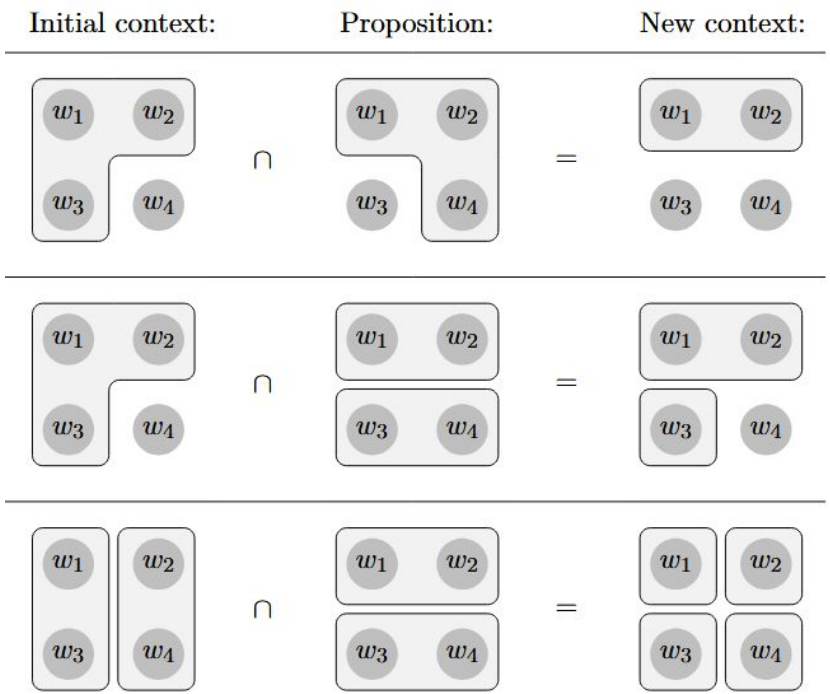
Inquisitive semantics: Contexts

- Context update C[P] := C ∩ P



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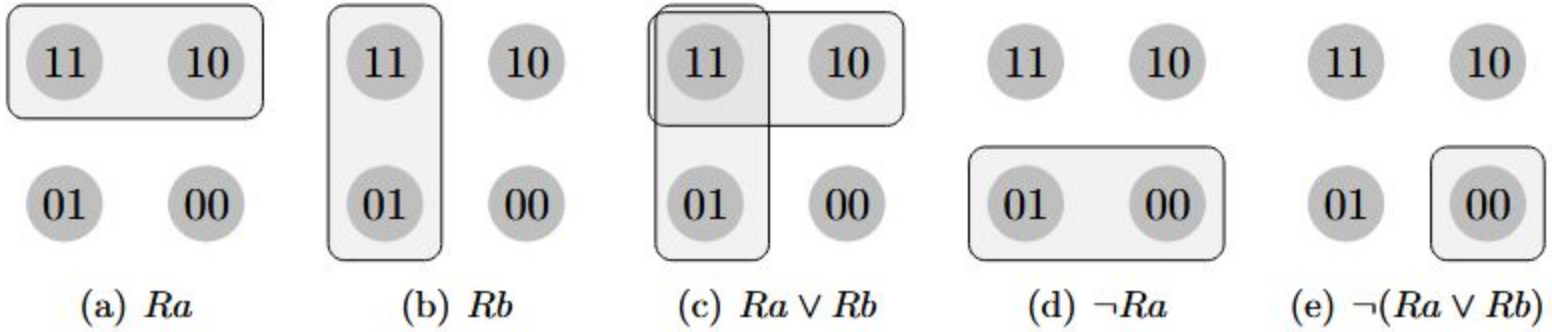
Inquisitive semantics: Basic operations and projection operators

- Let's assume that our language contains just one unary predicate symbol, R, and two individual constants, a and b.
- The domain of discourse consists of just two objects, denoted by a and b.
- Our logical space consists of four worlds, one in which both Ra and Rb are true, one in which Ra is true but Rb is false, one in which Rb is true but Ra is false, and one in which neither Ra nor Rb is true. These worlds will be labeled 11, 10, 01, and 00, respectively.
- For concreteness, we will read Ra as 'Ann is in Rome', and Rb as 'Bob is in Rome'.



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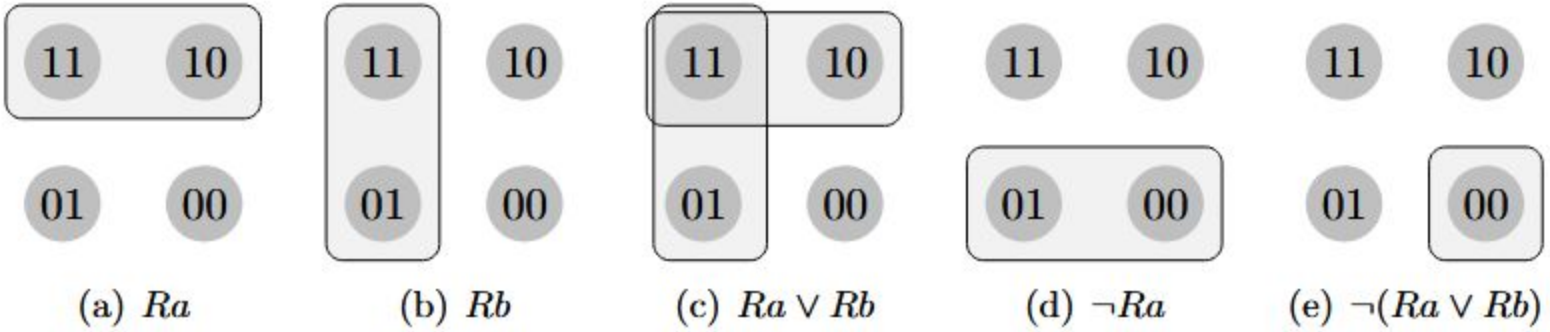
Inquisitive semantics: Basic operations and projection operators

- Let us first consider the proposition expressed by the atomic sentences Ra and Rb.
- [Ra] consists of all states s such that every world in s makes Ra true, i.e., the state {11, 10} and all substates thereof.
- Ra is non-inquisitive. It provides the information that Ann is in Rome, and it does not request any further information.
- Analogously, Rb provides the information that Bob is in Rome, without requesting any further information.



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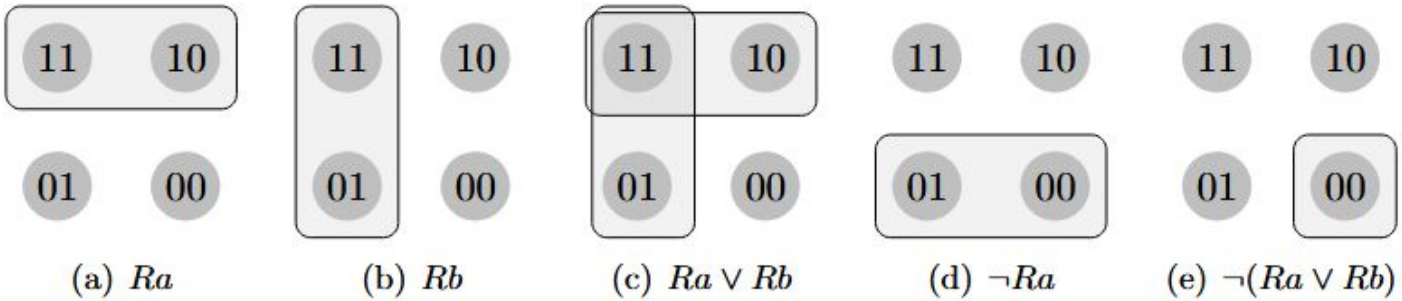
Inquisitive semantics: Basic operations and projection operators

- Disjunction [Ra ∨ Rb] consists of those states that are in [Ra] or in [Rb]. These are {11, 10}, {11, 01}, and all substates thereof.
- Since info(Ra ∨ Rb) = ∪[Ra ∨ Rb] = {11, 10, 01} ≠ W , the disjunction Ra ∨ Rb is informative. It provides the information that at least one of Ann and Bob is in Rome.
- Since there are two maximal elements, it is also inquisitive. In order to settle the issue that it raises, one has to establish either that Ann is in Rome, or that Bob is in Rome.



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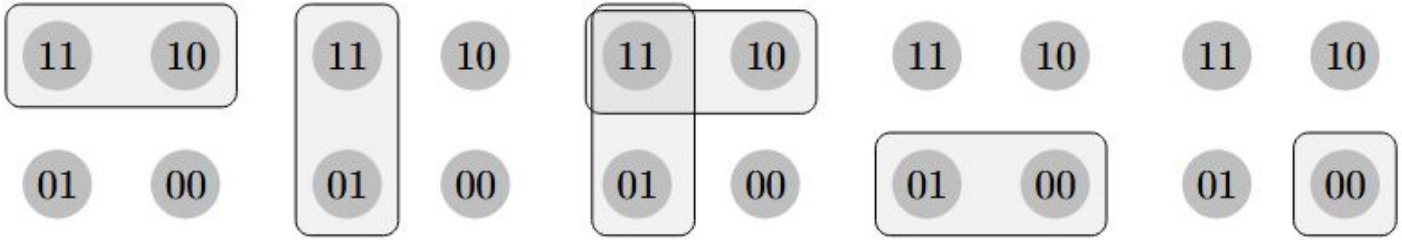
Inquisitive semantics: Basic operations and projection operators

- A disjunctive sentence Ra ∨ Rb does not necessarily correspond to a disjunctive declarative sentence in English like (1) below, or to a disjunctive interrogative sentence like (2).

  (1) Ann is in Rome or Bob is in Rome. !(Ra ∨ Rb)

  (2) Is Ann in Rome, or is Bob in Rome? ?(Ra ∨ Rb) or Ra ∨ Rb, depending on intonation



(a) Ra (b) Rb (c) $Ra \vee Rb$ (d) $\neg Ra$ (e) $\neg(Ra \vee Rb)$

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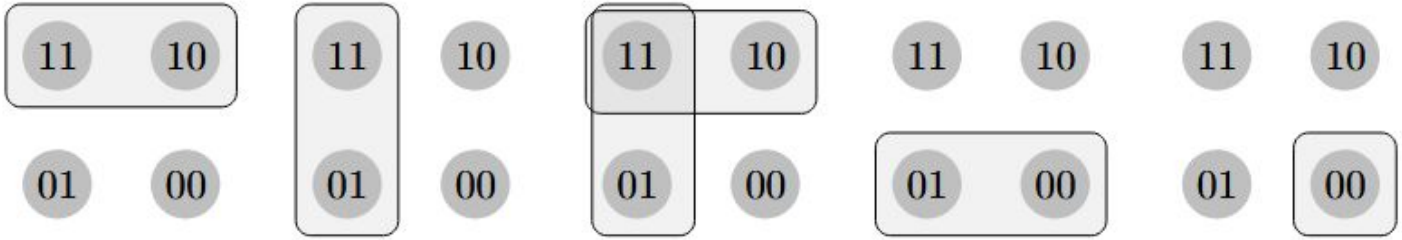
Inquisitive semantics: Basic operations and projection operators

- [¬Ra] consists of all states s such that s does not have any world in common with any state in [Ra]. Thus, [¬Ra] consists of all states that do not contain the worlds 11 and 10, which are |¬Ra| = {01, 00} and all substates thereof.
- ¬Ra is non-inquisitive. It provides the information that Ann is not in Rome, and does not request any further information.



(a) Ra (b) Rb (c) $Ra \vee Rb$ (d) $\neg Ra$ (e) $\neg(Ra \vee Rb)$

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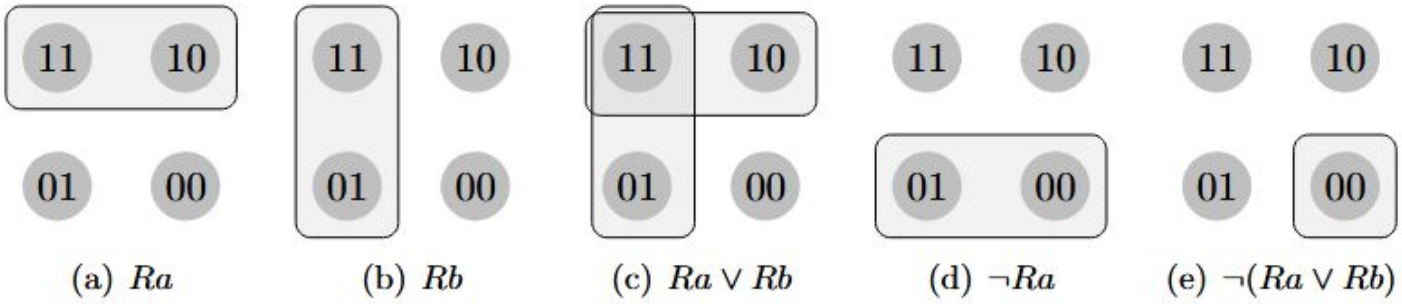
Inquisitive semantics: Basic operations and projection operators

- [¬(Ra ∨ Rb)] consists of all states which do not have a world in common with any state in [Ra ∨ Rb]. Thus, [¬(Ra ∨ Rb)] consists of all states that do not contain the worlds 11, 10, and 01, which are {00}and ∅.
- ¬(Ra ∨ Rb) provides the information that neither Ann nor Bob is in Rome and does not request any further information.
- The absolute pseudo-complement of a proposition always contains a greatest element.
- This means that a negative sentence ¬ϕ is never inquisitive; it simply provides the information that ϕ is false, and does not request any further information.



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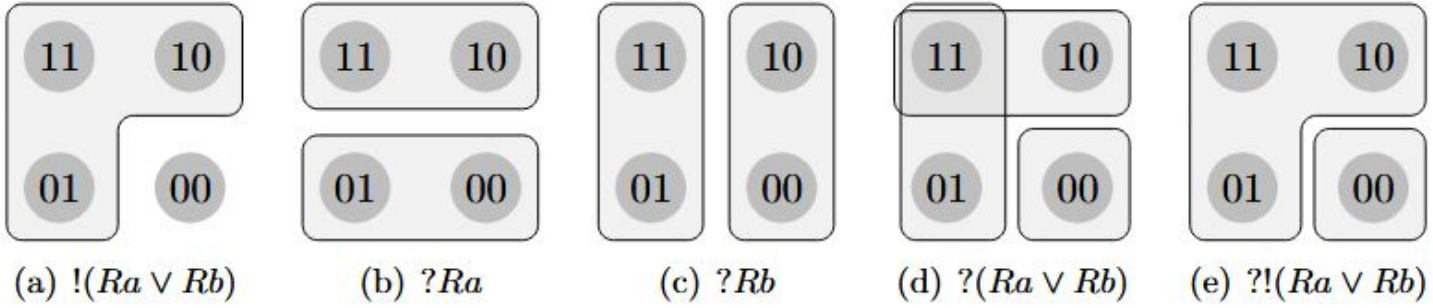
Inquisitive semantics: Basic operations and projection operators

- Let us consider !(Ra ∨ Rb), which abbreviates ¬¬(Ra ∨ Rb).
- For any sentence ϕ, !ϕ is a non-inquisitive proposition with the same informative content as ϕ.



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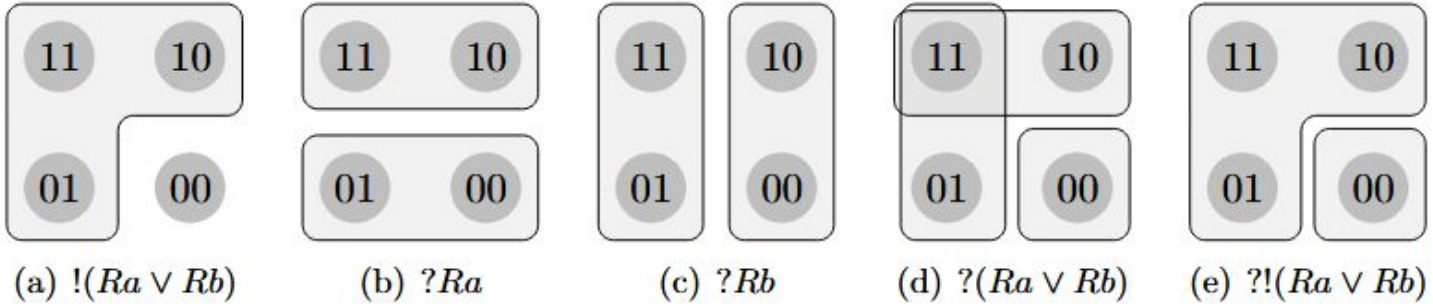
Inquisitive semantics: Basic operations and projection operators

- Consider ?Ra, which is an abbreviation of Ra ∨ ¬Ra.
- [?Ra] = [Ra ∨ ¬Ra] consists of all states that are either in [Ra] or in [¬Ra]. These states are |Ra|, |¬Ra|, and all substates thereof.
- Since info(?Ra) = W , ?Ra is non-informative.
- Since [?Ra] contains two alternatives, it is inquisitive. In order to settle the issue that it raises, one has to establish either that Ann is in Rome, or that Ann is not in Rome. That is, one has to establish whether Ann is in Rome.



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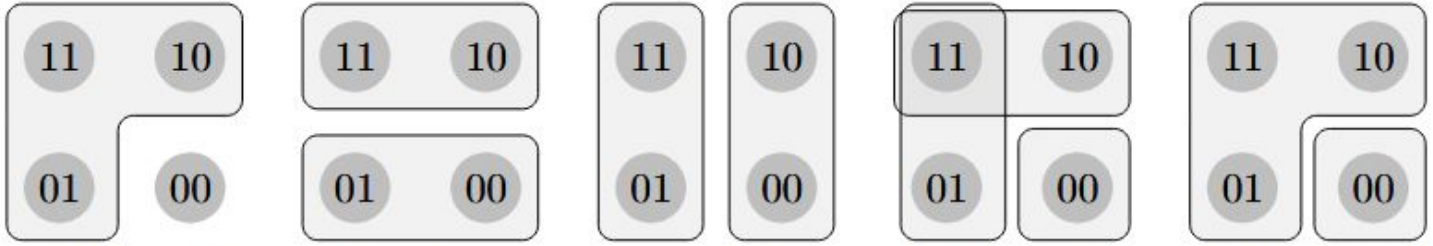
Inquisitive semantics: Basic operations and projection operators

- If ? applies to the disjunction Ra ∨ Rb, which is already inquisitive, then it yields the proposition depicted in (d).
- [Ra ∨ Rb] already contains two alternatives, |Ra| and |Rb|; ? adds a third alternative, which is the set of worlds that are neither in |Ra| nor in |Rb|.
- Thus, in order to resolve the issue raised by ?(Ra ∨ Rb), one has to establish either that Ann is in Rome, or that Bob is, or that neither is.



(a) $!(Ra \vee Rb)$ (b) $?Ra$ (c) $?Rb$ (d) $?(Ra \vee Rb)$ (e) $?!(Ra \vee Rb)$

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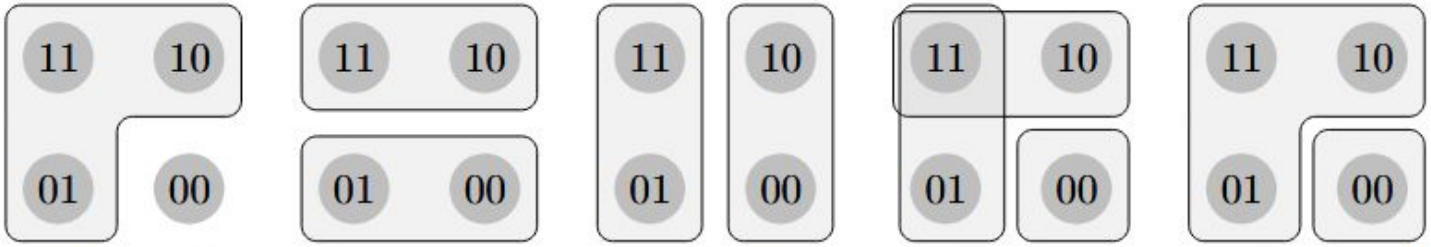
Inquisitive semantics: Basic operations and projection operators

- Let us consider a case where ! and ? both apply, one after the other: ?!(Ra ∨ Rb).
- [!(Ra ∨ Rb)] contains a single alternative, consisting of all worlds where at least one of a and b is in Rome.
- ? adds a second alternative, which is the set of worlds where neither Ann nor Bob is in Rome.
- Notice that the resulting proposition differs from that expressed by ?(Ra ∨ Rb), which contains three alternatives rather than two. In order to settle the issue expressed by ?!(Ra ∨ Rb) it is sufficient to establish that at least one of Ann and Bob is in Rome.
- In order to settle the issue expressed by ?(Ra ∨ Rb) this is not sufficient; rather, it needs to be established of one of Ann and Bob that he or she is in Rome, or that neither of them is.



(a) $!(Ra \vee Rb)$ (b) $?Ra$ (c) $?Rb$ (d) $?(Ra \vee Rb)$ (e) $?!(Ra \vee Rb)$

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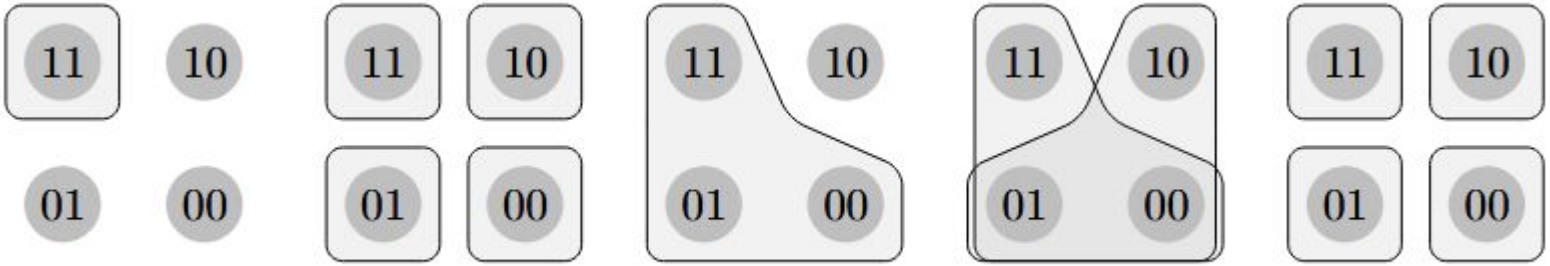
Inquisitive semantics: Basic operations and projection operators

- [Ra ∧ Rb] consists of those states that are both in [Ra] and in [Rb]. These are {11} and ∅. Thus, [Ra ∧ Rb] has a greatest element, namely {11}, and accordingly Ra ∧ Rb provides the information that both Ann and Bob are in Rome and does not request any further information.



(a) $Ra \wedge Rb$ (b) $?Ra \wedge ?Rb$ (c) $Ra \rightarrow Rb$ (d) $Ra \rightarrow ?Rb$ (e) $\forall x.?Rx$

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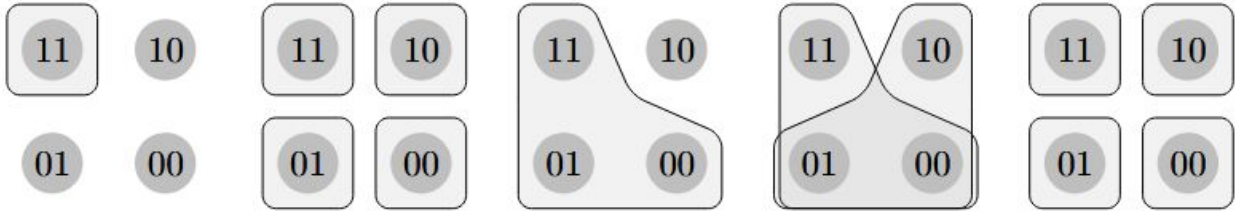
Inquisitive semantics: Basic operations and projection operators

- The proposition [?Ra ∧ ?Rb] contains four alternatives, |Ra ∧ Rb|, |Ra ∧ ¬Rb|, |¬Ra ∧ Rb|, and |¬Ra ∧ ¬Rb|.
- Since these alternatives together cover the entire logical space ?Ra ∧ ?Rb is non-informative.
- Since there is more than one alternative, ?Ra ∧ ?Rb is inquisitive.
- Thus, the conjunction is a purely inquisitive sentence which requests enough information to settle both the issue whether Ann is in Rome, contributed by ?Ra, and the issue whether Bob is in Rome, contributed by ?Rb.
- If ϕ and ψ are non-inquisitive, then the conjunction ϕ ∧ ψ is non-inquisitive as well, conveying both the information provided by ϕ and the information provided by ψ; on the other hand, if ϕ and ψ are non-informative, then the conjunction ϕ∧ψ is non-informative as well, expressing an issue which is settled just in case both the issue expressed by ϕ and the one expressed by ψ are settled.



(a) $Ra \wedge Rb$ (b) $?Ra \wedge ?Rb$ (c) $Ra \rightarrow Rb$ (d) $Ra \rightarrow ?Rb$ (e) $\forall x.?Rx$

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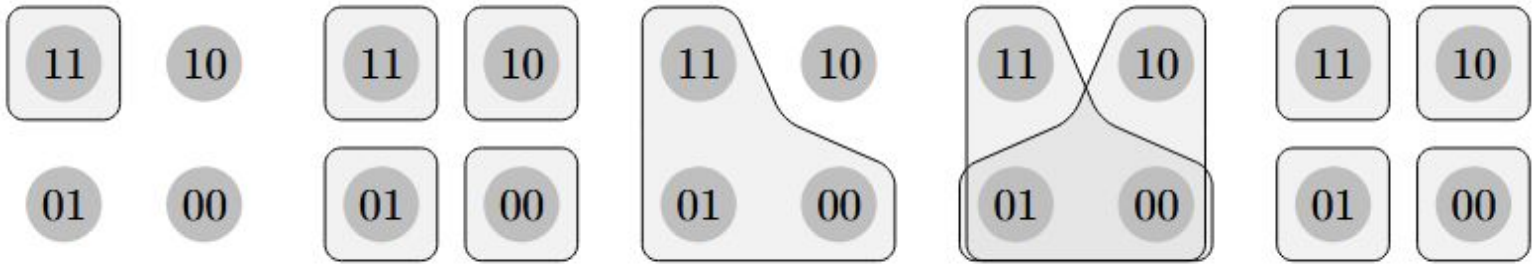
Inquisitive semantics: Basic operations and projection operators

- Implication [Ra → Rb] consists of all states s such that every substate t ⊆ s that is in [Ra] is also in [Rb]. These are all and only those states that are contained in |Ra → Rb| = {11, 01, 00}.
- [Ra → Rb] has a unique greatest element, which means that the implication Ra → Rb is a non-inquisitive sentence which, provides the information that if Ann is in Rome, then so is Bob.



(a) $Ra \wedge Rb$ (b) $?Ra \wedge ?Rb$ (c) $Ra \rightarrow Rb$ (d) $Ra \rightarrow ?Rb$ (e) $\forall x.?Rx$

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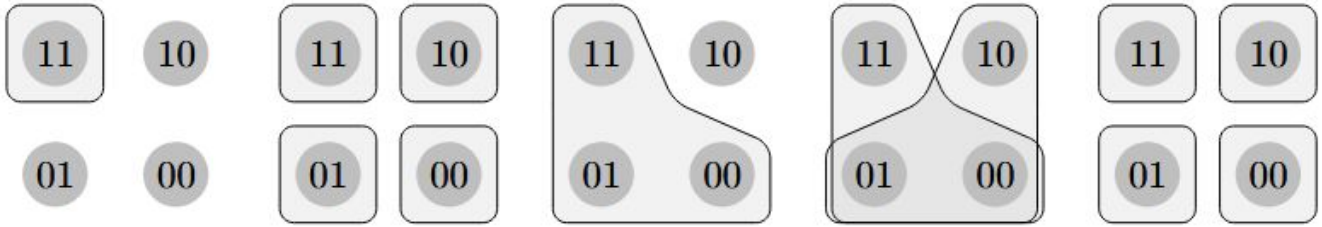
Inquisitive semantics: Basic operations and projection operators

- The proposition [Ra → ?Rb] contains two alternatives, |Ra → Rb| = {11, 01, 00}, and |Ra → ¬Rb| = {10, 01, 00}. Since these two alternatives together cover the entire logical space, our implication is non-informative. Since there is more than one alternative, the implication is inquisitive. In order to settle the issue that it raises, one must either establish Ra → Rb, or Ra → ¬Rb. In the former case one establishes that if Ann is in Rome, then so is Bob; in the latter case, that if Ann is in Rome, then Bob isn't. So Ra → ?Rb requests enough information to establish whether Bob is in Rome under the assumption that Ann is.
- If ψ is non-inquisitive, then ϕ → ψ is non-inquisitive as well; similarly, if ψ non-informative, then ϕ → ψ is non-informative as well.



(a) $Ra \wedge Rb$ (b) $?Ra \wedge ?Rb$ (c) $Ra \rightarrow Rb$ (d) $Ra \rightarrow ?Rb$ (e) $\forall x.?Rx$

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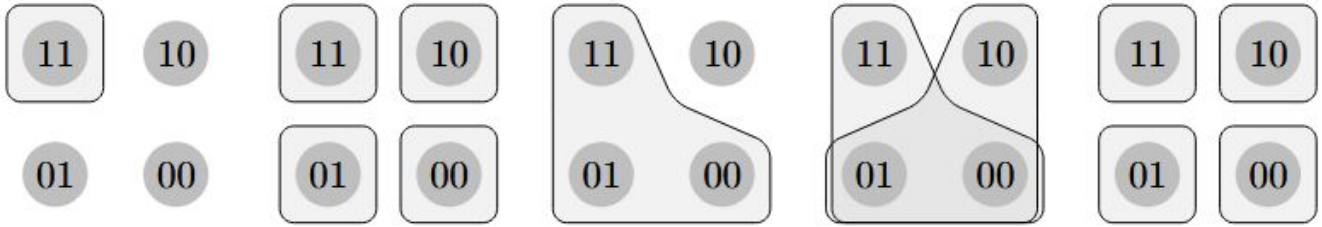
Inquisitive semantics: Basic operations and projection operators

- Existential quantification behaves essentially like disjunction and universal quantification behaves essentially like conjunction.
- Since our current domain of discourse consists of only two objects, denoted by a and b, $\exists$x.Rx expresses exactly the same proposition as Ra $\vee$ Rb, and $\forall$x.Rx expresses exactly the same proposition as Ra $\wedge$ Rb.
- Consider the proposition expressed by $\forall$x.?Rx.
- Notice that this proposition induces a partition on the logical space, where each block of the partition consists of worlds that agree on the extension of R. Thus, $\forall$x.?Rx asks for an exhaustive specification of the individuals that are in Rome.



(a) $Ra \wedge Rb$ (b) $?Ra \wedge ?Rb$ (c) $Ra \rightarrow Rb$ (d) $Ra \rightarrow ?Rb$ (e) $\forall x.?Rx$

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Inquisitive semantics: Sufficient conditions for non-inquisitivity

- Atomic sentences are always non-inquisitive;
- $\neg\phi$ is always non-inquisitive;
- $!\phi$ is always non-inquisitive;
- If ϕ and ψ are non-inquisitive, then so is $\phi \wedge \psi$;
- If ψ is a non-inquisitive, then so is $\phi \rightarrow \psi$ for any antecedent ϕ;
- If $\phi(d')$ is a non-inquisitive for all $d \in D$, then so is $\forall x\phi(x)$.
- Any sentence that does not contain $\vee$, $\exists$, or ? is non-inquisitive.

- In many natural languages, question words are homophonous with words for disjunction and/or existentials.

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Inquisitive semantics: Sufficient conditions for non-informativity

- $?\phi$ is always non-informative;
- If ϕ and ψ are non-informative, so is $\phi \wedge \psi$;
- If ψ is non-informative, then so are $\phi \vee \psi$ and $\phi \rightarrow \psi$, for any ϕ;
- If $\phi(d')$ is non-informative for all $d \in D$, then so is $\forall x\phi(x)$;
- If $\phi(d')$ is non-informative for some $d \in D$, then so is $\exists x\phi(x)$.

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